

AI Alignment Research Fellowship | AI Safety Global Society

Project Report Summary

Title: Embedding and Detecting Hidden Signals in Informal / Low-Resource Language (Pidgin English) Using Large Language Models

Background

Steganography is the practice of hiding secret information inside ordinary communication. With the rise of large language models (LLMs), there is a growing concern that they can be misused to embed covert signals in natural text. While most existing research focuses on high-resource languages such as English, little attention has been given to informal or low-resource languages like Pidgin English.

Objectives

This project explored whether LLMs can be prompted to embed hidden binary signals in Pidgin-to-English translations and whether such signals can be detected with simple classifiers. The key goals were to:

1. Develop 16 rule-based encoding methods for Pidgin steganography.
2. Compare three prompting strategies: loose, strict, and subtle.
3. Evaluate detectability using TF-IDF vectorization with logistic regression.

Methodology

- Data Generation: TinyLlama-1.1B-Chat was used to generate Pidgin-English datasets under different prompt styles.
- Experiments:
 - Loose prompts (32 samples) → low compliance, poor detection.
 - Strict prompts (320 samples) → near-perfect compliance, 92% detection accuracy.
 - Subtle prompts (250 samples) → natural outputs, perfect (100%) detection.
 - Large-scale dataset (984 samples) → 100% compliance, 100% detection.
- Detection: Applied TF-IDF (unigrams + bigrams) with logistic regression to classify bit-0 vs bit-1 encodings.

Key Findings

- LLMs can reliably embed covert signals in Pidgin text with the right prompting.
- Loose prompting is unreliable, strict prompting enforces compliance but produces rigid text, and subtle prompting achieves a balance of naturalness and detectability.
- Even subtle encodings were detected with 100% accuracy using simple statistical methods.

Contributions

- First systematic study of steganography in Pidgin English.
- Open-source dataset generation pipeline with prompts, datasets, and code.
- Comparative evaluation of prompting strategies for steganography.
- Insights into AI safety risks in low-resource language settings.

Future Directions

- Extend to other low-resource and multilingual contexts.
- Explore neural steganography methods e.g paraphrasing, adversarial encoding.
- Investigate defensive strategies for detecting covert misuse of LLMs.